

OCR Further Mathematics
Pure Core - Further Vectors
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) Points A , B and C have coordinates $(4, 2, 0)$, $(1, 5, 3)$ and $(1, 4, -2)$ respectively. The line l passes through A and B .

Find a cartesian equation for l .

[3]

- (b) M is the point on l that is closest to C .

Find the coordinates of M .

[4]

- (c) Find the exact area of the triangle ABC .

[4]

2(a)

The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$.

The plane Π has equation $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -5 \\ -3 \end{pmatrix} = 4$.

Find the position vector of the point of intersection of l_1 and Π .

[3]

- (b) Find the acute angle between l_1 and Π .

[3]

- (c) A is the point on l_1 where $\lambda = 1$.

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 l_2 is the line with the following properties.

- ♦ l_2 passes through A
- ♦ l_2 is perpendicular to l_1
- ♦ l_2 is parallel to Π

Find, in vector form, the equation of l_2 .

[3]

3(a)

A plane P has the equation $\mathbf{r} \cdot \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix} = 15$. The point C is the point $(4, -5, 1)$.

Find the shortest distance between l and C .

[3]

(b) Lines l_1 and l_2 have the following equations.

$$l_1: \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

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© Find, in exact form, the distance between l_1 and l_2

[5]

4(a) The equations of two intersecting lines l_1 and l_2 are

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ a \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} \quad l_2: \mathbf{r} = \begin{pmatrix} 7 \\ 9 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

where a is a constant.

The equation of the plane Π is

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix} = -14.$$

l_1 and Π intersect at Q .

l_2 and Π intersect at R .

Verify that the coordinates of R are $(13, 3, -14)$.

[2]

(b) Determine the exact value of the length of QR .

[7]



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- 5 The equations of two non-intersecting lines, l_1 and l_2 , are

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \quad l_2: \mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}.$$

Find the shortest distance between lines l_1 and l_2 .

[5]



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- 6 The equation of a plane is $4x + 2y + z = 7$.
The point A has coordinates $(9, 6, 1)$ and the point B is the reflection of A in the plane.

Find the coordinates of the point B .

[6]



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7 Points A , B and C have coordinates $(0, 1, -4)$, $(1, 1, -2)$ and $(3, 2, 5)$ respectively.

(a) Find the vector product $\overrightarrow{AB} \times \overrightarrow{AC}$.

[3]

(b) Hence find the equation of the plane ABC in the form $ax+by+cz = d$.

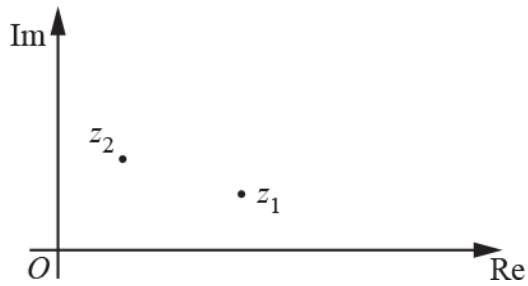
[2]

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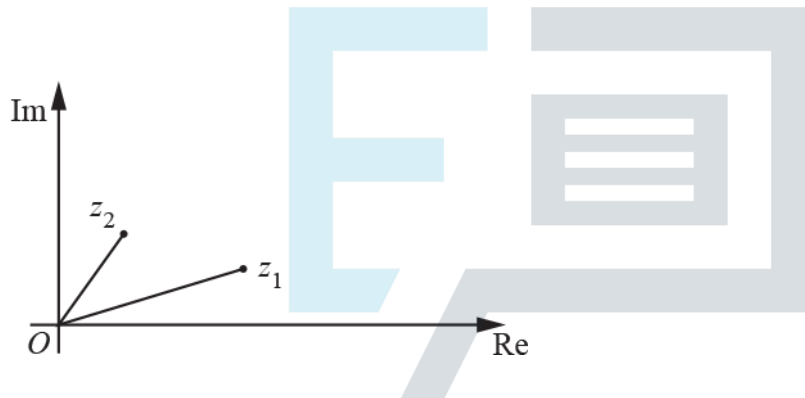
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- 8 (a) The Argand diagram below shows the two points which represent two complex numbers, z_1 and z_2 .



On the copy of the diagram below

- draw an appropriate shape to illustrate the geometrical effect of adding z_1 and z_2 ,
- indicate with a cross (x) the location of the point representing the complex number $z_1 + z_2$. [2]



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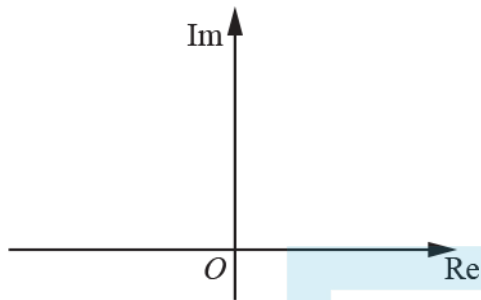
You are given that $\arg z_3 = \frac{1}{4}\pi$ and $\arg z_4 = \frac{3}{8}\pi$.

(b)

In each part, sketch and label the points representing the numbers z_3 , z_4 and z_3z_4 on the diagram below. You should join each point to the origin with a straight line.

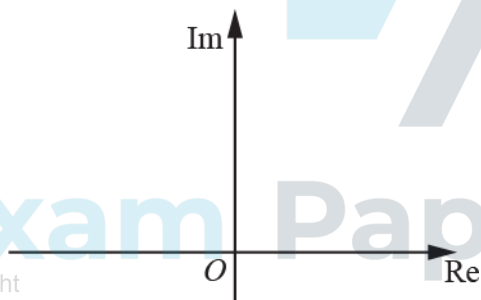
(i) $|z_3| = 1.5$ and $|z_4| = 1.2$

[2]



(ii) $|z_3| = 0.7$ and $|z_4| = 0.5$

[2]



9 Two planes, Π_1 and Π_2 , have equations $3x + 2y + z = 4$ and $2x + y + z = 3$ respectively.

(a) Find the acute angle between Π_1 and Π_2 .

[3]

The line L has equation $x = 1 - y = 2 - z$.

(b) Show that L lies in both planes.

[5]

10 The equation of a plane, Π , is

$$\Pi: \mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}.$$

(a) Find a vector which is perpendicular to Π .

[2]

(b) Hence find an equation for Π in the form $rx + y = p$.

[2]

(c) Find in the form $\sqrt{q}$ the shortest distance between Π and the origin, where q is a rational number.

[2]



11 Line l_1 has Cartesian equation

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$$l_1: \frac{-x}{2} = \frac{y-5}{2} = \frac{-z-6}{7}.$$

(a) Find a vector equation for l_1 .

[2]

Line l_2 has vector equation

$$l_2: \mathbf{r} = \begin{pmatrix} 2 \\ 7 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}.$$

(b) Find the point of intersection of l_1 and l_2 .

[3]

(c) Find the acute angle between l_1 and l_2 .

[3]

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The lines l_1 and l_2 have equations $\frac{x-7}{2} = \frac{y-1}{-1} = \frac{z-6}{3}$ and $\frac{x-2}{1} = \frac{y-6}{2} = \frac{z+2}{1}$ respectively.

(a) Show that l_1 and l_2 intersect.

[4]



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(b) Find the cartesian equation of the plane that contains l_1 and l_2 .

[3]

13 Plane Π has equation $3x - y + 2z = 33$. Line l has the following vector equation.

$$l: \quad \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$

(a) Find the acute angle between Π and l .

[3]

(b) Find the coordinates of the point of intersection of Π and l .

[3]

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(c) S is the point $(4, 5, -5)$. Find the shortest distance from S to Π .

[2]

14 The equations of two lines are

$$\mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \text{ and } \mathbf{r} = \begin{pmatrix} -1 \\ 8 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 1 \\ -5 \end{pmatrix}.$$

Find the coordinates of the point where these lines intersect.

[4]



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15 The plane Π and the line l have equations

$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} = 7 \text{ and } \mathbf{r} = \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

respectively. The point A has coordinates $(1, 2, -4)$.

(i) Find the shortest distance from the point A to the plane Π .

[3]

(ii) Find the acute angle between Π and l .

[3]

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(iii) Find the point where the line parallel to l passing through A intersects the plane Π .

[4]

16 A plane Π_1 passes through the points $(1, 2, -1)$, $(2, -3, 1)$ and $(-1, 0, 2)$.

- (i) Show that the plane Π_1 has equation $11x + 7y + 12z = 13$.

[4]

The plane Π_2 has equation $3x + y + z = 4$.

- (ii) Find a vector equation of the line of intersection of Π_1 and Π_2 .

[4]

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- (iii) Find the acute angle between Π_1 and Π_2 .

[2]

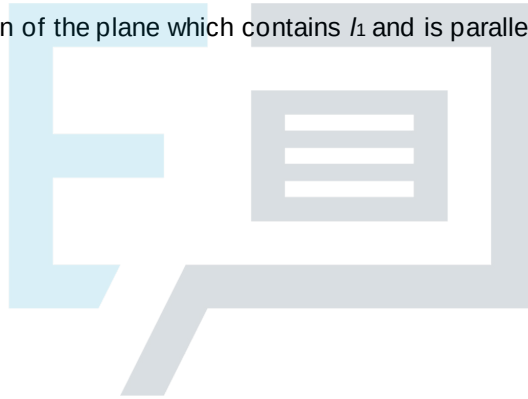
17 The lines l_1 and l_2 have equations $\frac{x-3}{1} = \frac{y-5}{2} = \frac{z+2}{-3}$ and $\frac{x-4}{2} = \frac{y+2}{-1} = \frac{z-7}{4}$.

(a) Find the shortest distance between l_1 and l_2 .

[5]

(b) Find a cartesian equation of the plane which contains l_1 and is parallel to l_2 .

[2]



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18 The equation of a plane Π is $x - 2y - z = 30$.

- (a) Find the acute angle between the line $\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 3 \\ 2 \end{pmatrix}$ and Π . [4]

- (b) Determine the geometrical relationship between the line $\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$ and Π . [4]

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19 The vector equations of two lines are as follows.

$$L: \mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix} + s \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \quad M: \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + t \begin{pmatrix} 5 \\ -3 \\ 1 \end{pmatrix}$$

(i) Show that the lines L and M meet, and find the coordinates of the point of intersection.

[4]



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- (ii) Show that the line L can also be represented by the equation $\mathbf{r} = \begin{pmatrix} 7 \\ 1 \\ 14 \end{pmatrix} + u \begin{pmatrix} -4 \\ 2 \\ -6 \end{pmatrix}$. [2]



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Find the shortest distance between the lines $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$.

21 The points A and B have position vectors relative to the origin O given by

[4]



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$$\overrightarrow{OA} = \begin{pmatrix} 3 \sin \alpha \\ 2 \cos \alpha \\ -1 \end{pmatrix} \text{ and } \overrightarrow{OB} = \begin{pmatrix} 2 \cos \alpha \\ 4 \sin \alpha \\ 3 \end{pmatrix},$$

where $0^\circ < \alpha < 90^\circ$. It is given that $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are perpendicular.

[5]

- (i) Calculate the two possible values of α .



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(ii) Calculate the area of triangle OAB for the smaller value of α from part (i).

[4]



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22 A triangle has vertices at $A(1, 1, 3)$, $B(5, 9, -5)$ and $C(6, 5, -4)$. P is the point on AB such that $AP : PB = 3 : 1$.

(i) Show that $\overrightarrow{CP}$ is perpendicular to $\overrightarrow{AB}$.

[4]



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(ii) Find the area of the triangle ABC .

[2]



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23 The plane Π passes through the points $(1, 2, 1)$, $(2, 3, 6)$ and $(4, -1, 2)$.

(i) Find a cartesian equation of the plane Π .

[5]

The line l has equation $\mathbf{r} = \begin{pmatrix} -1 \\ -2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}$.

(ii) Find the coordinates of the point of intersection of Π and l .

[3]

(iii) Find the acute angle between Π and l .

[3]

24 Find the shortest distance between the lines with equations

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-5}{-1} \quad \text{and} \quad \frac{x-3}{4} = \frac{y-1}{-2} = \frac{z+1}{3}.$$

[7]

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25 Two lines have equations

$$r = 3i + 5j - k + \lambda(2i + j + k) \text{ and } r = 4i + 10j + 19k + \mu(i - j + \alpha k),$$

where α is a constant.

Find the value of α in each of the following cases.

(i) The lines intersect at the point (7, 7, 1).

[3]



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(ii) The angle between their directions is 60° .

[4]



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(i) Find a vector equation of the line of intersection of the planes $2x + y - z = 4$ and $3x + 5y + 2z = 13$.

[4]

(ii) Find the exact distance of the point $(2, 5, -2)$ from the plane $2x + y - z = 4$.

[2]

27 The points $O(0, 0, 0)$, $A(2, 8, 2)$, $B(5, 5, 8)$ and $C(3, -3, 6)$ form a parallelogram $OABC$. Use a scalar product to find the acute angle between the diagonals of this parallelogram.

[5]

28 The equations of three lines are as follows.

Line A: $r = i + 4j + k + s(-i + 2j + 2k)$

Line B: $r = 2i + 8j + 2k + t(i + 3j + 5k)$

Line C: $r = -i + 19j + 15k + u(2i - 4j - 4k)$

(i) Show that lines A and B are skew.

[4]

(ii) Determine, giving reasons, the geometrical relationship between lines A and C.

[2]

29 The line l has equations $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-7}{5}$. The plane Π has equation $4x - y - z = 8$.

(i) Show that l is parallel to Π but does not lie in Π .

[3]



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(ii) The points $A(1, -2, 7)$ is on l . Write down a vector equation of the line through A which is perpendicular to l .
Hence find the position vector of the point on l which is closest to A .

[4]

(iii) Hence write down a vector equation of the line in l which is parallel to l and closest to it.

[1]



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30 The plane Π passes through the points with coordinates $(1, 6, 2)$, $(5, 2, 1)$ and $(1, 0, -2)$.

(i) Find a vector equation of Π in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$.

[2]

(ii) Find a cartesian equation of Π .

[4]

31 Determine whether the lines whose equations are

$$\mathbf{r} = (1 + 2\lambda)\mathbf{i} - \lambda\mathbf{j} + (3 + 5\lambda)\mathbf{k} \text{ and } \mathbf{r} = (\mu - 1)\mathbf{i} + (5 - \mu)\mathbf{j} + (2 - 5\mu)\mathbf{k}$$

are parallel, intersect or are skew.

[6]

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32

The plane Π has equation $x + 2y - 2z = 5$. The line l has equation $\frac{x-1}{2} = \frac{y+1}{5} = \frac{z-2}{1}$.

- (i) Find the coordinates of the point of intersection of l with the plane Π .

[3]



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(ii) Calculate the acute angle between l and Π .

[3]

(iii) Find the coordinates of the two points on the line l such that the distance of each point from the plane Π is 2.

[5]

33 Points $A(2, 2, 5)$, $B(1, -1, -4)$, $C(3, 3, 10)$ and $D(8, 6, 3)$ are the vertices of a pyramid with a triangular base.

(i) Calculate the lengths AB and AC , and the angle BAC .

[4]

(ii) Show that $\vec{AD}$ is perpendicular to both $\vec{AB}$ and $\vec{AC}$.

[3]

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(iii) Calculate the volume of the pyramid $ABCD$.

[3]

[The volume of the pyramid is $V = \frac{1}{3} \times \text{base area} \times \text{perpendicular height}$.]

34 The equations of two lines are

$$r = i + 2j + \lambda(2i + j + 3k) \text{ and } r = 6i + 8j + k + \mu(i + 4j - 5k).$$

(i) Show that these lines meet, and find the coordinates of the point of intersection.

[5]

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(ii) Find the acute angle between these lines.

[3]

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35 The points $A(3, 2, 1)$, $B(5, 4, -3)$, $C(3, 17, -4)$ and $D(1, 6, 3)$ form a quadrilateral $ABCD$.

(i) Show that $AB = AD$.

[2]

(ii) Find a vector equation of the line through A and the mid-point of B

[3]

(iii) Show that C lies on the line found in part (ii).

[1]

(iv) What type of quadrilateral is $ABCD$?

[1]

36 Two planes have equations

$$x + 2y + 5z = 12 \text{ and } 2x - y + 3z = 5.$$

(i) Find the acute angle between the planes.

[3]

(ii) Find a vector equation of the line of intersection of the planes.

[4]

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37 The lines l_1 and l_2 have equations

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \text{ and } \mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}$$

respectively.

(i) Find the shortest distance between the lines.

[5]

(ii) Find a cartesian equation of the plane which contains l_1 and which is parallel to l_2

[2]



END OF QUESTION PAPER

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